Steps to calculate **n** and **sigma** using calculation

1. STEP 1: To find power at different distance using below formula

$$P_r(d_i) = P_r(d_0) - 10n \log_{10}\left(\frac{d_i}{d_0}\right)$$

Where, i = 1 to 20 (for our experiment)

d_i = distance

$P_r(d_i) =$ power at distance di

$P_r(d_0) = 0$ dBm at $d_0 = 100$ m (assumption for our experiment)

So we will have $P_r(d_1)$ to $P_r(d_{20})$ in terms of n

$P_r(d_1) = 0 - 10n \log_{10}\left(\frac{d_1}{100}\right) = -10n\, log_{10}\left(\frac{450}{100}\right) = -6.53212n$

$P_r(d_2) = 0 - 10n\, log_{10}\left(\frac{d_2}{100}\right) = -6.43453n$

$P_r(d_3) = 0 - 10n\, log_{10}\left(\frac{d_3}{100}\right) = -6.33468n$

$P_r(d_4) = 0 - 10n\, log_{10}\left(\frac{d_4}{100}\right) = -6.23249n$

$P_r(d_5) = 0 - 10n\, log_{10}\left(\frac{d_5}{100}\right) = -6.12784n$

$P_r(d_6) = 0 - 10n\, log_{10}\left(\frac{d_6}{100}\right) = -6.02060n$

$P_r(d_7) = 0 - 10n\, log_{10}\left(\frac{d_7}{100}\right) = -5.91065n$

$P_r(d_8) = 0 - 10n\, log_{10}\left(\frac{d_8}{100}\right) = -5.79784n$

$P_r(d_9) = 0 - 10n\, log_{10}\left(\frac{d_9}{100}\right) = -5.68202n$

$P_r(d_{10}) = 0 - 10n\, log_{10}\left(\frac{d_{10}}{100}\right) = -5.56303n$

$P_r(d_{11}) = 0 - 10n\, log_{10}\left(\frac{d_{11}}{100}\right) = -5.44068n$

$P_r(d_{12}) = 0 - 10n\, log_{10}\left(\frac{d_{12}}{100}\right) = -5.31479n$

$P_r(d_{13}) = 0 - 10n\, log_{10}\left(\frac{d_{13}}{100}\right) = -5.18514n$

$P_r(d_{14}) = 0 - 10n\, log_{10}\left(\frac{d_{14}}{100}\right) = -5.05150n$

$P_r(d_{15}) = 0 - 10n\, log_{10}\left(\frac{d_{15}}{100}\right) = -4.91362n$

$P_r(d_{16}) = 0 - 10n\, log_{10}\left(\frac{d_{16}}{100}\right) = -4.77121n$

$$P_r(d_{17}) = 0 - 10n\,log_{10}\left(\frac{d_{17}}{100}\right) = -4.62398n$$

$$P_r(d_{18}) = 0 - 10n\,log_{10}\left(\frac{d_{18}}{100}\right) = -\,4.47158n$$

$$P_r(d_{19}) = 0 - 10n\,log_{10}\left(\frac{d_{19}}{100}\right) = -4.31364n$$

$$P_r(d_{20}) = 0 - 10n\,log_{10}\left(\frac{d_{20}}{100}\right) = -\,4.14973n$$

Here d_1 to d_{20} is from experiment

2. Now calculate Minimum mean square error (MMSE) and n
   MMSE is J(n)

$$J(n) = \sum_{i=1}^{k} \left(\left(P_r - \widehat{P_r}\right)\right)^2$$

Where,

P_r is the computed value.

$\widehat{P_r}$ is the calculated value.

computed value is from experiment
calculated is from above step 1.

Here $J(n)$ is calculated as follows :-

$$\begin{aligned} J(n) = {} & (-39.62 + 6.53213n)^2 \\ & +(-33.27 + 6.43453n)^2 \\ & +(-46.49 + 6.33468n)^2 \\ & +(-24.70 + 6.23249n)^2 \\ & +(-34.58 + 6.12784n)^2 \\ & +(-40.46 + 6.02060n)^2 \\ & +(-25.40 + 5.91065n)^2 \\ & +(-30.52 + 5.79784n)^2 \\ & +(-34.46 + 5.68202n)^2 \\ & +(-12.53 + 5.56303n)^2 \\ & +(-31.64 + 5.44068n)^2 \\ & +(-33.20 + 5.31479n)^2 \\ & +(-32.28 + 5.18514n)^2 \\ & +(-25.58 + 5.05150n)^2 \\ & +(-21.78 + 4.91362n)^2 \\ & +(-35.38 + 4.77121n)^2 \\ & +(-22.26 + 4.62398n)^2 \\ & +(-30.97 + 4.47158n)^2 \\ & +(-33.19 + 4.31364n)^2 \\ & +(-23.42 + 4.14973n)^2 \end{aligned}$$

Use the below formula on each square

$$(a + bn)^2 = a^2 + 2abn + b^2n^2$$

so we get finally $J(n)$ as

$$J(n) = 602.98861n^2 - 6744.37195n + 19832.45650$$

Differentiate with respect to n:

$$\frac{dJ(n)}{dn} = 2(602.98861)n - 6744.37195$$

$$\frac{dJ(n)}{dn} = 1205.97722n - 6744.37195$$

$$\boxed{\frac{dJ(n)}{dn} = 0}$$

$$1205.97722n - 6744.37195 = 0$$

$$n = \frac{6744.37195}{1205.97722}$$

$$n = 5.59245$$

Now $J(n)$ is $J(n) = 602.98861n^2 - 6744.37195n + 19832.45650$

$$J(5.59245) = 973.66215$$

3. Now find sigma using below formula

$$\sigma^2 = \frac{J(n)}{N}$$

$N = 20$

$$\sigma^2 = \frac{973.66215}{20}$$
$$\sigma^2 = 48.68311 \text{ dB}^2$$

$$\sigma = \sqrt{48.68311}$$
$$\sigma = 6.97733 \text{ dB}$$

(Note : - In exam, only 6 readings are enough for path loss fading. so, we will have i = 1 to 6. N = 6. Rest Entire procedure remains the same. This all above calculation we have to do on paper with calculator)

Frequency reuse

Just use this formula to verify value of N

$$N = i^2 + ij + j^2$$

for us i=2 and j=1 . therefore $N = 2^2 + 2X1 + 1^2$=7